

HOW CAN ACADEMIC SPIN-OFFS IMPROVE THEIR MARKET PERFORMANCE BY TAKING INTO CONSIDERATION THE VOICE OF THEIR CUSTOMERS?

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Abstract

Purpose – The research objective is to demonstrate that Kano with HWWP model can improve the market performance of academic spin-offs (AS) by translating the voice of the customer into design priorities.

Methodology/approach – The paper presents the methodological steps for customer needs assessment on a AS case study from Romania. A questionnaire with 12 customer features was applied on 64 clients. Because the features of the product are agglomerated in the quandrant, the the uniformity test of the HWWP model is applied in order to show which product characteristic should be improved first.

Findings – The paper provides new insights for academic spin-offs in Romania by developing new strategies in market orientation in order survive.

Research limitations/implications – The limitation is given by applying the method on a single case study.

Practical implications – The study can provide guidance's for universities, research institutes in order to integrate this method in the training courses held for researchers, students.

Originality/value – The novelty of the paper is given by Kano and HWWP methodology which can improve the survival of the academic spin-off, helping the innovative venture to succed on the market, by adapting the product to the customer needs.

Key words: aacademic spin-off, market orientation, HWWP model

Introduction

Market orientation has been long considered as a key factor for business success (DeLuca et al., 2010) since it involves the concept of delivering superior value to the customer (Han et al., 1998). A definition of market orientation is given by Slater and Narver (1994) which involves three dimensions: customer orientation, competitor orientation and inter-functional coordination. Han et al. (1998) showed that market orientation facilitates an organization's innovativeness, which, in turn, positively influences its business performance.

Buratti et al. (2021) argue the relevance of being market oriented when exploiting new innovative technologies for commercial profit, highlighting the specific case of university spin-offs or how they are also called, academic spin-offs (AS). But what are these ASs and what is the value they provide to the current marketplace? Academic spin-offs are marking the future of sustainable regional development (Vac and Fitiiu, 2017) because they have a unique opportunity to market R&D outcomes from university labs into real use. Their creation is encouraged by governmental spheres (especially in the European Union) since they represent a very important technological transfer mechanism from scientific research towards product design and final commercialization, increasing regional competitiveness (Mihali et al., 2022).

But despite being „a driver of local economic development, their growth is slow, especially for companies based in the EU” (Buratti et al., 2021). Hesse and Sternberg (2017) consider as a main cause for their performance limitations the way they are managed. As they suggest, these spin-offs are run more as research laboratories. Also, their founders take decisions based on prejudices or feelings rather than on

objective data from potential customers (Robets, 1990) and are obsessed to prove their scientific technology than to meet market needs. Rasmussen et al. (2011) also conclude that academics, despite being innovation driven in their research domain, they lack the skills to identify opportunities in the market context. The consequences are failure to use the added value for commercializing technological innovations or failure to direct attention towards potential customer needs (Würmseher, 2017). Moreover, in a study on Eastern European countries with statistical research on Romanian ASs, Mihali et al. (2022) suggest that the higher the number of academics in an academic spin-off, the higher the failure to succeed in the market due to the same lack of market orientation.

Thus, market orientation of the academic founder positively affects the business (Grandi and Grimaldi, 2005) and might be a source of competitive advantage over time (Buratti et al., 2021). If not acquired, the lack of market knowledge is an obstacle for growth (Van Geenhuizen and Soetano, 2009).

The present paper aims to provide a methodology for academic spin-offs to force them change their perspective from research towards their customer needs. The first step is to focus on product features and understand customers' needs regarding them. Because academic entrepreneurs in a AS are primarily researchers with technical background, providing them with a methodological model is key for its acceptance. Therefore, the Kano model with its refinement – the HWWP (Health Weapon Wealth Prospect) model has been selected for introducing customer orientation. Based on Pugna et al. (2021)'s elasticity curves in the HWWP model, the academic entrepreneur has all the necessary data to understand the voice of the customer and direct the focus on the commercial context.

In the next section the Kano and HWWP model are explained together with methodological steps for customer orientation, followed by a case study of a Romanian academic spin-off which used this methodology to develop its products in accordance with market requirements. Finally, conclusions are outlined.

The Kano model and the HWWP model

Despite its foundation more than two decades ago, the Kano model is an important tool for assessing the voice of the customer for new offers or for improving existing ones. Kano et al. (1984) envisages to explain the roles each product or service feature play for potential or current customers. With the help of a specific Kano questionnaire, each feature is evaluated as belonging to one of six different Kano categories, each with its specificity and strategic potential: attractive, one-dimensional, must-be, indifferent, reverse and questionable.

Because the Kano model had some disadvantages largely discussed by scholars, new refinements appeared. One of them is the HWWP (Health Weapon Wealth Prospect) model, proposed by Potra et al. (2017) for new innovative products or services. With its four dimensions based on customer satisfaction and importance ratings variables, with suggestive names like: Health Weapon Wealth and Prospect, each feature or customer requirement is visually allocated for strategic decision making (as can be seen in Figure 1. The initial HWWP model).

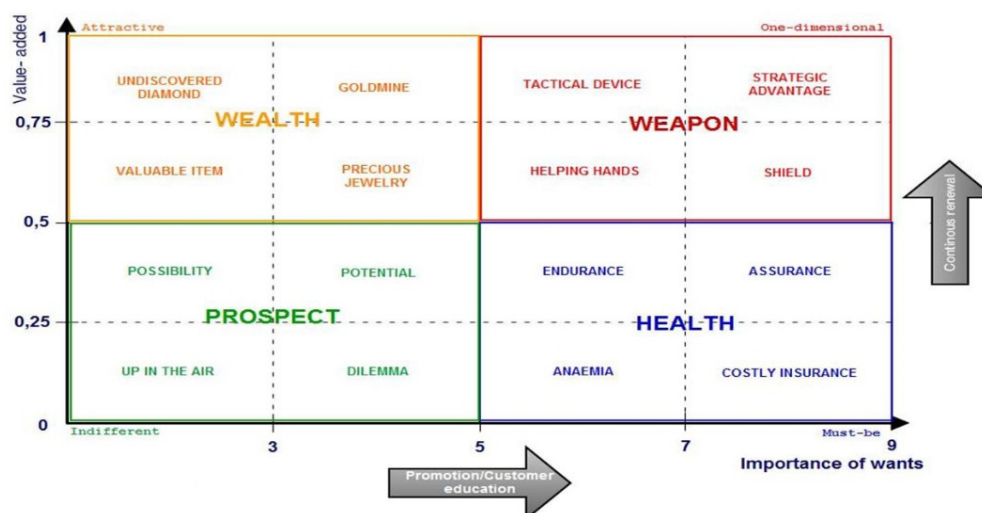


Figure 1. The initial HWWP model (Source: Potra et al., 2017)

The four dimensions have each four subdomains in which customer requirements can be found for a thorough differentiation between them. Nevertheless, the HWWP has been also improved by Pugna et al. (2021) with the help of elasticity curves when the linear model is not an option. In this way, the lifecycle of all features can be strategically assessed for managerial decision-making.

Proposed methodology for customer orientation

In Figure 2 the methodology of the present paper is presented. The steps academic entrepreneurs need to follow are explained and offer guidelines for changing their perspective towards their customer needs with the help of a sound and proven model.

As first phases, the product or service academics want to analyze or improve needs to be selected. Existing and new functionalities or also called features (customer requirements/quality attributes) need to be delineated because managers have to assess the existing functionalities added value and importance for existing customer but also propose new functionalities for product improvement which will trigger customer delight.

A basic knowledge of the Kano and Importance questionnaire is necessary for their adequate construction. All information is provided in literature (Sauerwein et al., 1996) for the construction and evaluation of questionnaire responses.

For the HWWP visual representation, Potra et al. (2017) present the way in which the two variables are computed. In Figure 2 presents the methodology steps for customer needs assessment.

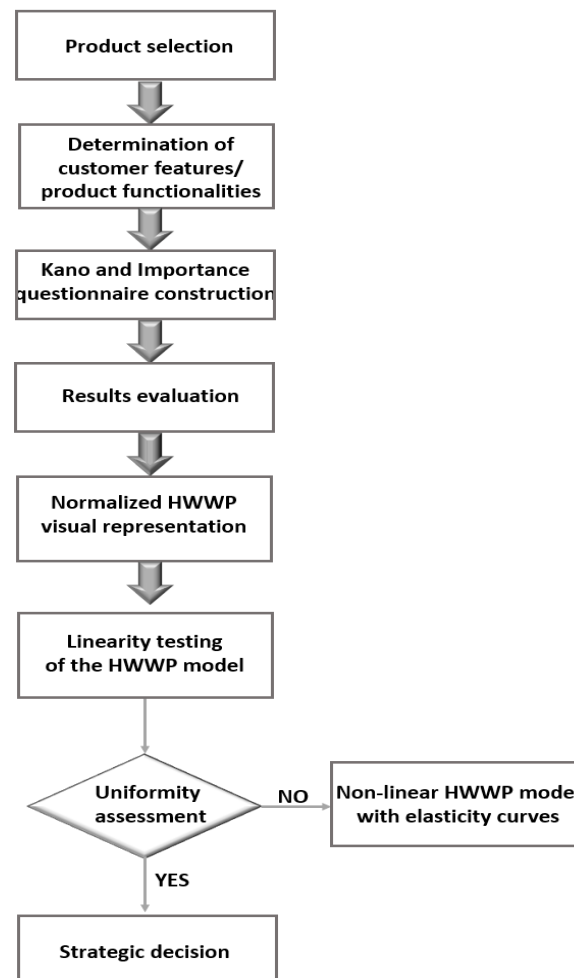


Figure 2. Methodology steps for customer needs assessment

If the methodology seems difficult to understand without statistical knowledge, the AS management team can stop at the fifth step when visually representing customer requirements in the HWWP classical

model. The linearity testing provides only a more thorough viewpoint in special cases when two or several features are very close to one another in a subdomain of the HWWP model. By using a statistical linearity test like Rayleigh or Chen and Hu test, the management team can assess if they need a non-linear HWWP model or the data is relevant enough in the linear one. Based on this last data, a strategic decision can be adequately taken.

To prove the methodology is not difficult to implement in an academic spin-off, the next chapter proposes a case study analysis.

Proofing the methodology works for academic spin-offs. A Case study from the Western part of Romania

The selected spin-off is a company active in the renewable energies industry from the Western part of Romania. Established in 2017 by a professor in the Timis county, it produces pallets and lighters. Financial details can be seen in Table 1.

Table 1. AS1 details from Timis county (Source: www.mfinante.ro)

Firm age (year)	Size (employees)	Market performance (turnover)
1 (2017)	1	0
2 (2018)	4	0
3 (2019)	3	155.000
4 (2020)	0	90.200
5 (2021)	2	10.000

The first step in improving its products has been to select the product they wanted to focus upon, namely the lighters. Then, for the determination of customer features, the authors have discussed with the company managers and delineated 12 functionalities presented in Table 2 for potential improvement purposes.

Table 2. Improvement functionalities for lighters and their description

Nr. crt.	Customer features/functionalities	Description
1	Recyclable polyethylene packaging	Customers usually desire recyclable packaging and the polyethylene maintains a correct humidity degree of the product
2	Wood imitation	The unrooted lighters are light colored and the rooted ones are darker. For burning in the fireplace, they are visually more appealing in a color which imitates wood
3	Rectangle shape of the product	
4	Calorific power	A higher calorific power is more demanded
5	Nice fir smell	When burning, such lighters will give off a pleasant smell of fir (the classical version has no smell or even a bad smell)
6	No dirt/ash	After burning the lighter in the fireplace or in the central heating device, the client wishes the ash level to be low to avoid additional dirt
7	Resistant	A better resistance of the product is desired, this means the lighters should remain in one piece and not be easily breakable
8	Small package	Lighters could be bought in smaller packages of 5 pieces or more
9	Home delivery	The products could have the home delivery option when ordered
10	Buy back option	At 5 kg of customer waste (corn cobs, straw...) 1 kg of briquettes is received
11	Card payment option	When paying the order, the customer could have the possibility to pay cash but also by card
12	Delivery in 24h	After ordering the products, the customer should expect them to be delivered in one day (very fast for the current situation)

For the third step, the authors have constructed a Kano and Importance questionnaire totaling a number of 24 questions (12 questions with functional and dysfunctional form for the Kano questionnaire and 12 importance assessment questions). After discussing with 64 clients and completing the questionnaires, the evaluation of results can be seen in Table 3.

Table 3. The Kano and importance questionnaire results

no.	Functionality	A	O	M	I	R	Q	Total	Kano category	Importance	Potential added value (CS)
1	Recyclable packaging	13	8	5	35	2	1	64	I	0.43	0.34
2	Wood imitation	29	5	1	22	1	6	64	A	0.47	0.59
3	Rectangle shape	7	5	2	32	11	7	64	I	0.39	0.25
4	Calorific power	22	26	6	8	0	2	64	O	0.73	0.77
5	Nice fir smell	31	14	1	13	2	3	64	A	0.48	0.76
6	No dirt/ash	4	35	9	12	0	4	64	O	0.71	0.65
7	Resistant	12	26	14	9	1	2	64	O	0.72	0.62
8	Small package	18	11	4	22	7	2	64	I-A	0.47	0.52
9	Home delivery	33	16	2	10	1	2	64	A	0.59	0.8
10	Buy back option	30	18	3	12	0	1	64	A	0.6	0.76
11	Card payment option	17	22	8	14	3	0	64	O-A	0.61	0.63
12	Delivery in 24h	26	20	3	13	2	0	64	A-O	0.57	0.74

The recyclable packaging together with the rectangle shape and small package options seem indifferent to customers. If they are not difficult to implement in the final product, the management team could try to include them in their final offer. The wood imitation, nice fir smell, home delivery and buy back options are attractive for customers, providing a nice surprise if met. At least two of them need to be implemented for a competitive advantage. But the caloric power, no dirt/ash, resistance and card payment options are one-dimensional, expected features customer expect to receive. Thus, these functionalities need not only to be implemented but the company has to perform them at the highest standards to win against competition.

The normalized HWWP visual model in Figure 3 presents all the 12 functionalities as they have been computed by using the voice of the 64 customers in one of the four quadrants.

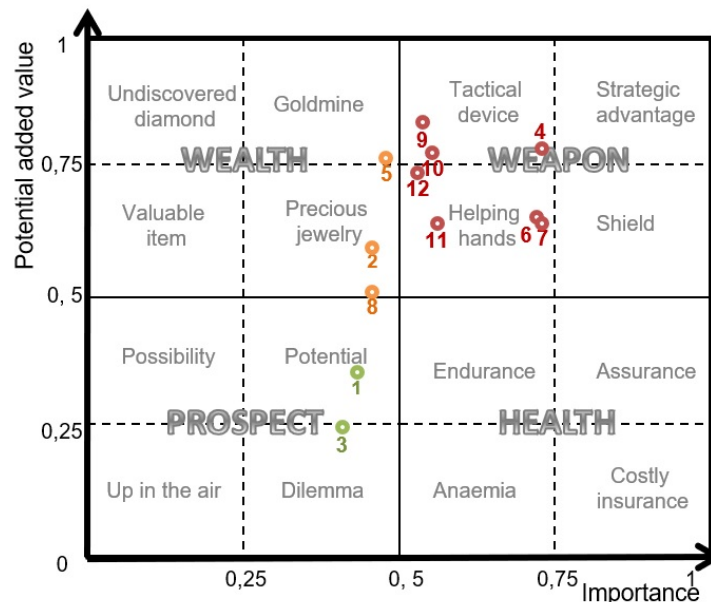


Figure 3. The normalized HWWP model for the 12 lighter proposed functionalities

Due to the fact that the points associated with the 12 functionalities are agglomerated in the weapon quadrant, we need to test the uniformity of the HWWP model as argued by Pugna et al. (2021) with

the help of the Rayleigh or Chen and Hu test. In the current case study we have chosen to use Rayleigh test. The computed value of the test is $11.82697 > 2.853$ (the value of $n=12$ from Table 4).

Table 4. The uniformity test for the HWWP model in the case of the 12 customer features

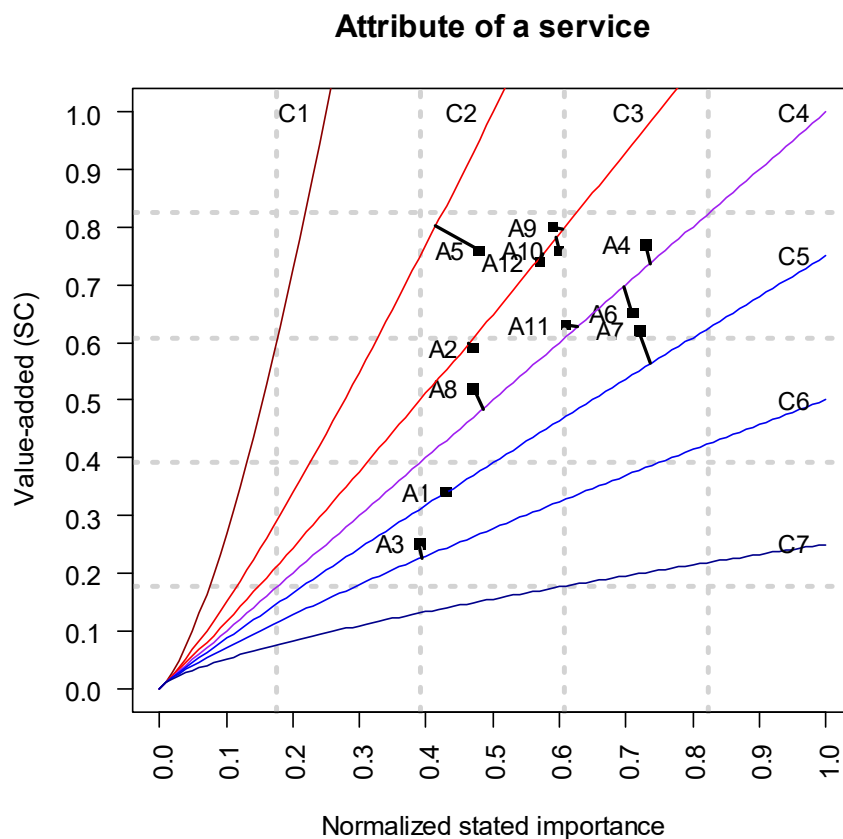
n	0.50	0.20	0.10	0.05	0.02	0.01	0.005	0.002	0.001
6	0.734	1.639	2.274	2.865	3.576	4.358	4.981	4.985	5.297
7	0.727	1.634	2.278	2.845	3.627	4.343	4.917	5.181	5.556
8	0.723	1.631	2.281	2.849	3.665	4.305	4.710	5.322	5.743
9	0.719	1.628	2.283	2.850	3.694	4.252	4.780	5.430	5.885
10	0.717	1.626	2.285	2.919	3.716	4.289	4.835	5.514	5.996
11	0.715	1.625	2.287	2.926	3.735	4.319	4.879	5.582	6.035
12	0.713	1.623	2.288	2.934	3.750	4.344	4.916	5.638	6.158
13	0.711	1.622	2.289	2.937	3.763	4.365	4.947	5.685	6.219
14	0.710	1.621	2.290	2.941	3.774	4.383	4.973	5.725	6.271
15	0.709	1.620	2.291	2.945	3.784	4.398	4.996	5.759	6.316
16	0.708	1.620	2.292	2.948	3.792	4.412	5.015	5.789	6.354
17	0.707	1.619	2.292	2.951	3.799	4.423	5.033	5.815	6.388
18	0.706	1.619	2.293	2.954	3.806	4.434	5.049	5.838	6.418
19	0.705	1.618	2.293	2.956	3.811	4.443	5.061	5.858	6.445
20	0.705	1.618	2.294	2.958	3.816	4.451	5.074	5.877	6.469
21	0.704	1.617	2.294	2.960	3.821	4.459	5.085	5.895	6.491
22	0.704	1.617	2.295	2.961	3.825	4.466	5.095	5.908	6.510
23	0.703	1.616	2.295	2.963	3.829	4.472	5.104	5.922	6.528
24	0.703	1.616	2.295	2.964	3.833	4.478	5.112	5.935	6.544
25	0.702	1.616	2.296	2.966	3.836	4.483	5.120	5.946	6.559
26	0.702	1.616	2.296	2.967	3.839	4.488	5.127	5.957	6.573
27	0.702	1.615	2.296	2.968	3.842	4.492	5.133	5.965	6.586
28	0.701	1.615	2.296	2.969	3.844	4.496	5.139	5.975	6.598
29	0.701	1.615	2.297	2.970	3.847	4.499	5.145	5.984	6.609
30	0.701	1.615	2.297	2.971	3.849	4.504	5.150	5.992	6.619
32	0.700	1.614	2.297	2.972	3.853	4.510	5.159	6.006	6.637
34	0.700	1.614	2.297	2.974	3.856	4.516	5.168	6.018	6.654
36	0.700	1.614	2.298	2.975	3.859	4.521	5.175	6.030	6.668
38	0.699	1.614	2.298	2.976	3.862	4.525	5.182	6.039	6.681
40	0.699	1.613	2.298	2.977	3.865	4.529	5.188	6.048	6.692
42	0.699	1.613	2.298	2.978	3.867	4.533	5.193	6.056	6.703
44	0.698	1.613	2.299	2.979	3.869	4.536	5.198	6.064	6.712
46	0.698	1.613	2.299	2.979	3.871	4.539	5.202	6.070	6.721
48	0.698	1.613	2.299	2.980	3.873	4.542	5.206	6.076	6.729
50	0.698	1.613	2.299	2.981	3.874	4.545	5.210	6.082	6.736
55	0.697	1.612	2.299	2.982	3.878	4.550	5.218	6.094	6.752
60	0.697	1.612	2.300	2.983	3.881	4.555	5.225	6.104	6.765
65	0.697	1.612	2.300	2.984	3.883	4.559	5.231	6.113	6.776
70	0.696	1.612	2.300	2.985	3.885	4.562	5.235	6.120	6.786
75	0.696	1.612	2.300	2.986	3.887	4.565	5.240	6.127	6.794
80	0.696	1.611	2.300	2.986	3.889	4.567	5.243	6.132	6.801
90	0.695	1.611	2.301	2.987	3.891	4.572	5.249	6.141	6.813
100	0.695	1.611	2.301	2.988	3.893	4.575	5.254	6.149	6.822
120	0.695	1.611	2.301	2.990	3.896	4.580	5.262	6.160	6.837
140	0.695	1.611	2.301	2.990	3.899	4.584	5.267	6.168	6.847
160	0.695	1.610	2.301	2.991	3.900	4.586	5.271	6.174	6.855
180	0.694	1.610	2.302	2.992	3.902	4.588	5.274	6.178	6.861
200	0.694	1.610	2.302	2.992	3.903	4.590	5.276	6.182	6.865
300	0.694	1.610	2.302	2.993	3.906	4.595	5.284	6.195	6.879
500	0.694	1.610	2.302	2.994	3.908	4.599	5.290	6.201	6.891
∞	0.6931	1.6094	2.3016	2.9957	3.9120	4.6052	5.2983	6.2146	6.9078

Discussion and conclusions

In conclusion, the uniformity hypothesis is rejected and a generalized HWWP model with elasticity curves is recommended (Pugna et al., 2021). The new approach offers a supplementary partition with 7 meridian elasticity curves (Figure 4) to assess the potential value each customer feature will provide to the analyzed product. The term elasticity is related to the sensitivity to customer demands. The greater the elasticity of a functionality, the more value it provides to the product and the company. Our two variables, namely the potential added value and the importance, are divided by a diagonal (a point of unit elasticity- curve 4) equal to 1. All data above the diagonal is considered as elastic and the below data as inelastic. As expressed already by Pugna et al. (2021), "In the elastic region, any small change in a quality attribute positively or negatively influences customer satisfaction, unlike in the inelastic region, where a value improvement has a reduced influence on customer satisfaction."

The elasticity curves provide relevant information for decision making by taking into consideration the voice of the potential customer and translating it into prospective strategies. For example, the feature 6 (no dirt/ash) and 7 (resistant) are similar, both situated in the weapon- helping hands dimension, but the resistance of the lighter (point 7) is inelastic. This means that the producer does not need to invest

further in improving the resistance of the product because the customer will not appreciate those efforts. But the effort in feature 6 (no dirt/ash) will lead proportionality to customer satisfaction. The feature 5 (nice fir smell) can be seen as the single feature in the upper elastic region, meaning that the producer could reach customer delight if investing in its quality, a real strategic advantage on the market.



The case study proves the Kano with HWWP visual model is a very useful instrument in determining the features of the new products, which a spin-off eager to launch on the market.

Academic spin-offs have the opportunity to take research towards new levels of innovation if the academic entrepreneur understands the importance of market orientation and especially the focus on the voice of the customer.

The present paper proposes methodological steps researchers are comfortable with to change their focus from pure research towards understanding their market. For Romania, a country with an emerging economy it is crucial for spin-offs to improve their performance and survival rate. Universities and research centers should take the initiative and organize specific entrepreneurial courses regarding the market needs by integrating the Kano model.

By recognizing the value of technological transfer and the AS's potential for the regional development, more studies should be carried out pointing out the importance of market requirements for AS on more spin-offs.

References

Buratti, N., G. Profumo, and L. Persico

2021

"The impact of market orientation on university spin-off business performance." *Journal of International Entrepreneurship*, Vol.19, pp. 104-129.

De Luca, L.M., G. Verona, and S. Vicari

2010

"Market orientation and R&D effectiveness in high-technology firms: an empirical investigation in the biotechnology industry." *J Prod Innov Manag*, 27(3):299–320.

Grandi, A, and R. Grimaldi

2005

"Academics' organizational characteristics and the generation of successful business ideas." *Jl Bus Ventur* 20(6):821–845.

Han, J. K., N. Kim, and R. Srivastava

1998

"Market orientation and organizational performance: is innovation a missing link?", *Journal of Marketing*, 62(4), 30–45.

Hesse, N., and R. Sternberg

2017

"Alternative growth patterns of university spin-offs: why so many remain small?." *International Entrepreneurship and Management Journal* 13, 953–984.

Kano, N., N. Seraku, F. Takahashi, and S. Tsuji

1984

"Attractive quality and Must-be quality." *The Journal of the Japanese Society for Quality Control*, 39-48, April

Mihali, L.M, S. Potra, L.I. Dungan, R. Negrea, and A. Cioabla

2022

Key Factors of AS Performance in Emerging Central and Eastern European Countries: Evidence from Romania, Sustainability in press.

Potra S.A., M. Izvercian, A.P. Pugna, and J.J. Dahlgaard

2017

The HWWP, a refined IVA-Kano model for designing new delightful products or services, *Total quality management and business excellence*, 28(1-2), pp.104-117

Pugna, A. P., S. A. Potra, and R. Negrea

2021

"A strategic decision making tool for new product and service design", *Management Decision*, Vol. 59 No. 2, pp. 406-425.

Rasmussen, E., S. Mosey, and M. Wright

2011

"The Evolution of Entrepreneurial Competencies: A Longitudinal Study of Academic spin-off Venture Emergence." *Journal of Management Studie* 48, 1314–1345.

Robert, E. B.

1990

Evolving toward product and market-orientation: the early years of technology-based firms. *J Prod Innov Manag* 7(4): 274–287.

Sauerwein, E., F. Bailom, K. Matzler, and H.H. Hinterhuber

1996

"The Kano Model: How to Delight Your Customers" in: *Preprints Volume I of the IX. International Working Seminar on Production Economics*, Innsbruck/Igls/Austria, pp. 313 -327

Slater, S.F, and J.C. Narver

1994

"Does competitive environment moderate the market orientation-performance relationship?" *Journal of Marketing* 58(1):46–55.

Vac, C.S., and A. Fitiu

2017

"Building Sustainable Development through Technology Transfer in a Romanian University." *Sustainability* 9

Van Geenhuizen, M., and Soetanto D.P.

2009

"Academic spin-offs at different ages: a case study in search of key obstacles to growth." *Technovation* 29(10):671–681.

Würmseher, M.

2017

"To each his own: matching different entrepreneurial models to the academic scientist's individual needs." *Technovation* 59:1–17